THE KOREAN ASTRONOMICAL SOCIETY
한국천문학회

2026 KAS Fall Meeting

An Agentic Vision-Language Model Pipeline for Zero-Shot Gravitational Wave Glitch Classification

Yubin Kim[1], Lanz Anthonee Lagman[2], Johnrob Bantang[3], Reinabelle Reyes [4]

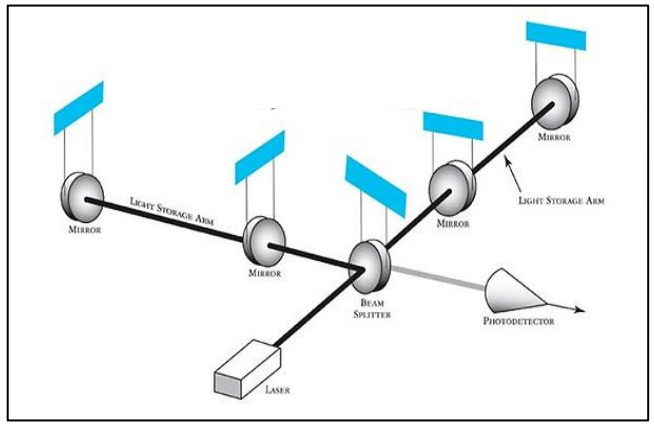
1. National Institute for Mathematical Sciences, 2. Computational Science Research Center, University of the Philippines Diliman, 3. National Institute of Physics, University of the Philippines Diliman, 4. Space Science Missions Bureau, Philippines Space Agency

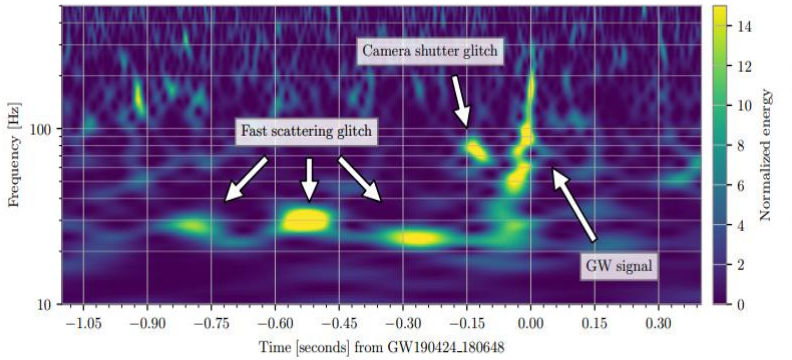
NIMS National Institute for Mathematical Sciences | South Korea

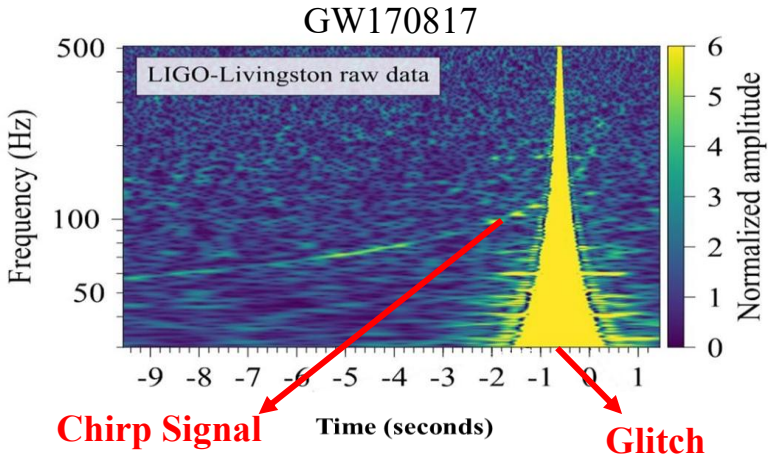
Overview

1. Introduction - Gravitational Wave Glitch Classification
2. VLM Integration to Astrophysics
3. Multi Agent Approach
4. Classification Performance
5. Conclusion & Summary

Gravitational Wave Glitch Classification







What is a Glitch?

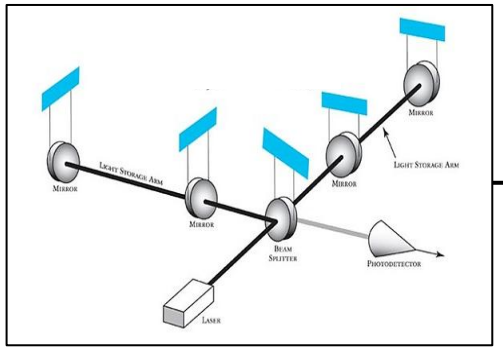
Non-astrophysical noise transients in ground-based GW detectors caused by environmental, instrumental, or unknown sources

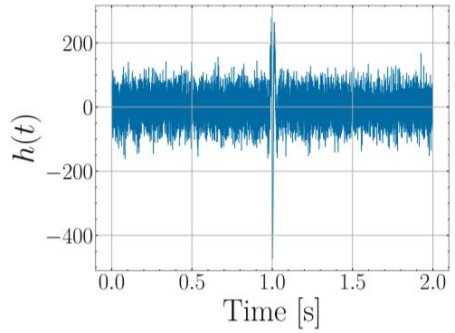
Why Does Glitch Classification Matter?

Glitches can mimic real signals and trigger false alarms (~1/3 of O3 public alerts were retracted), overlap with genuine signals, bias parameter estimation, and degrade overall detector data quality.

- A guide to LIGO-Virgo detector noise and extraction of transient gravitational-wave signals, The LIGO Scientific Collaboration And The Virgo Collaboration, 2020
- GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run, LVK Collaboration, 2023

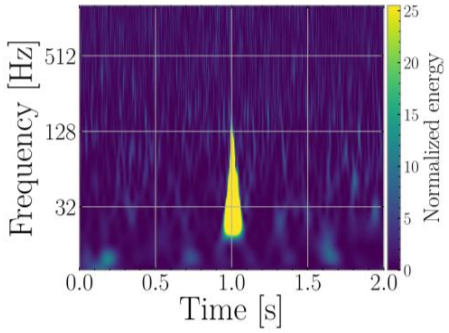
Glitch Identification and Classification



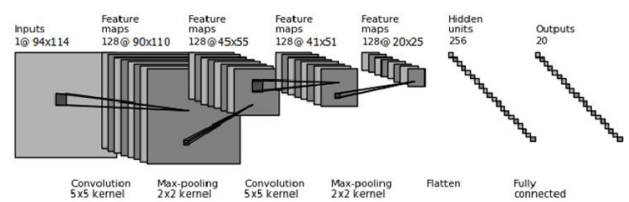


Strain Data (Timeseries)

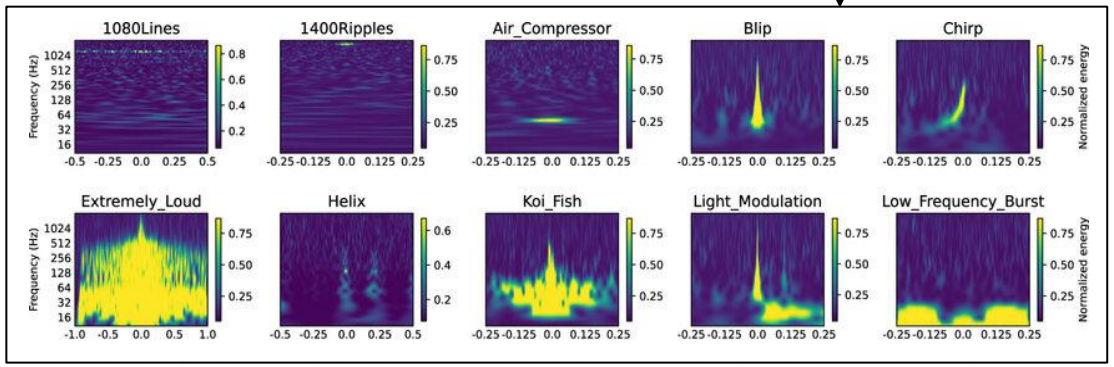
Omicron & GWpy



2D Spectrogram Image



Machine Learning Image Classification



Gravity Spy Dataset: LIGO O1,O2

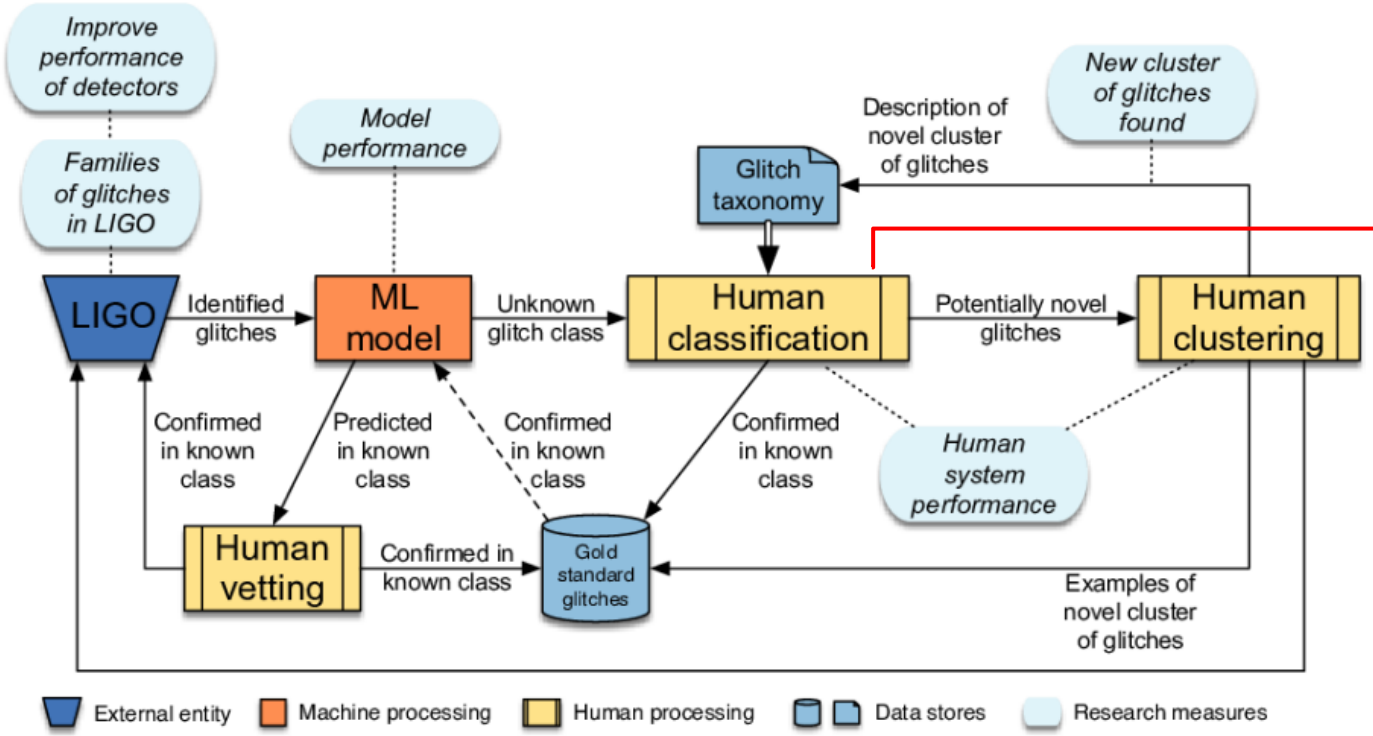
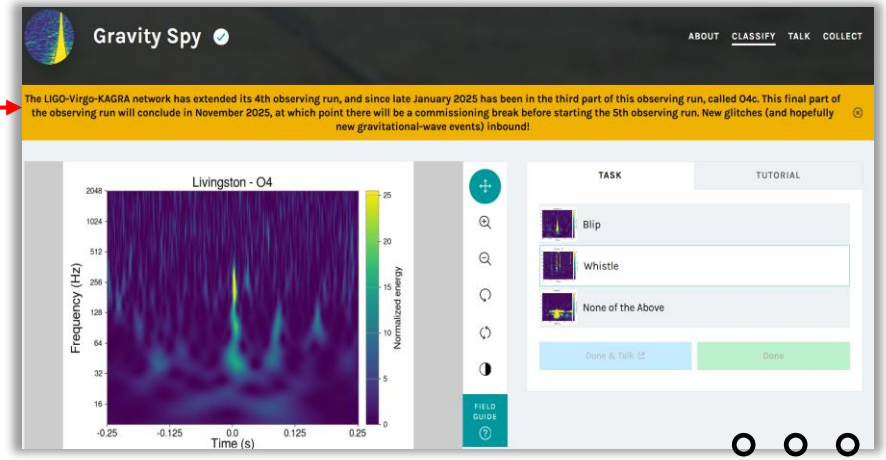


Figure 2: Gravity Spy system architecture, and overall data flow through the interconnected, interdisciplinary components of the project.

Zooniverse citizen science platform





- Gravity Spy: integrating advanced LIGO detector characterization, machine learning, and citizen science, M Zevin et al 2017 Class. Quantum Grav. 34 064003
- Machine learning for Gravity Spy: Glitch classification and dataset , S. Bahaadini et al. / Information Sciences 4 4 4 (2018) 172–186

How do we replace this pipeline with Vision-Language Model?

Advantages of VLMs over Supervised Classifiers

Explainability



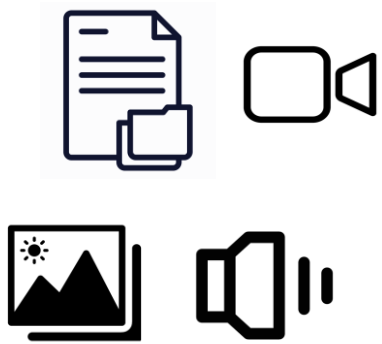
- VLMs can be used to generate natural-language descriptions of input images
- Supports explainable classification through auditable reasoning traces

Flexibility



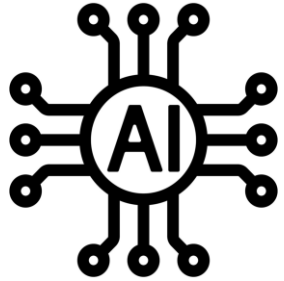
- Zero-shot inference eliminates the need for labeled training data and GPU retraining cycles.
- Classification taxonomy can be updated by simply modifying text prompts.

Multimodal Reasoning



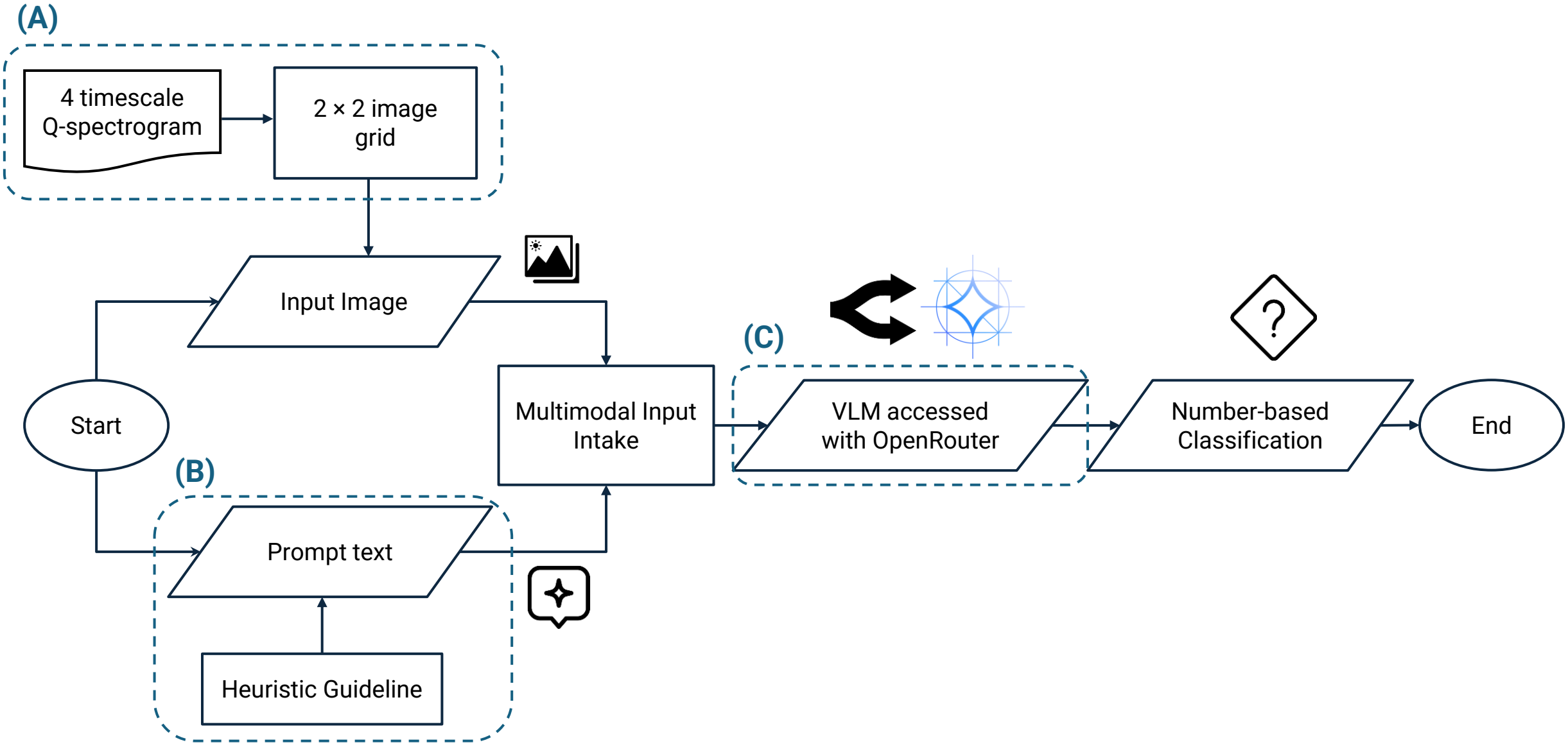
- Jointly processes image and metadata inputs within a single reasoning step.
- Replaces rigid branching architectures with a flexible, prompt-driven framework

Agentic Extension

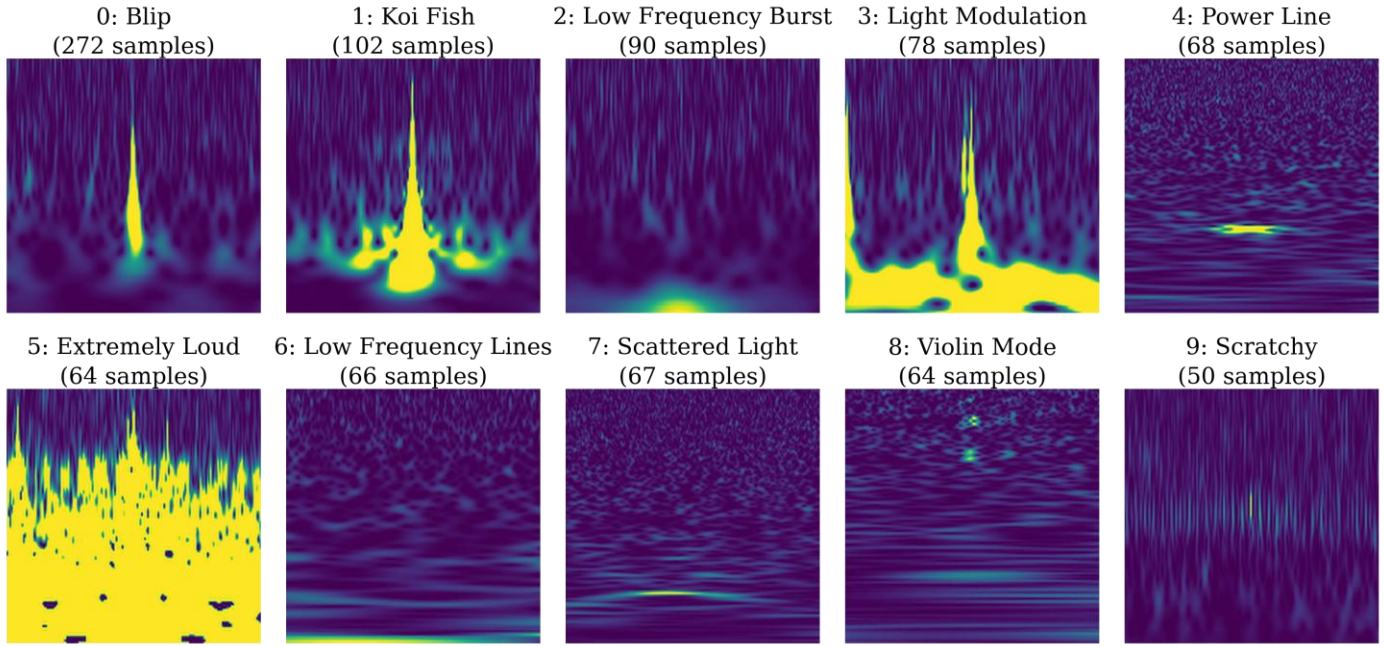


- Distributes classification into specialized sub-tasks across modular agent components
- Breaks down the classification pipeline into modular, manageable sub-steps.

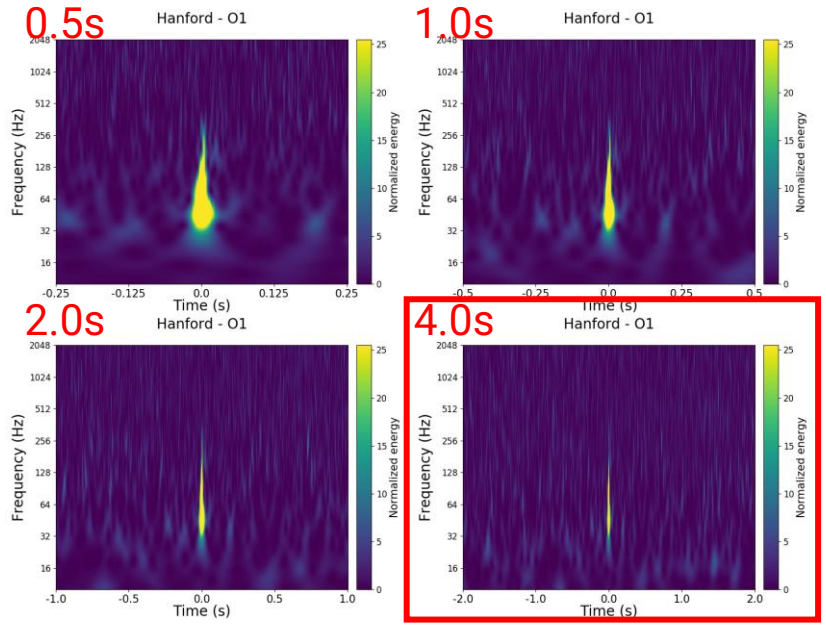
Classifier Agent Workflow



(A) Input Image Generation



1600×1200 RGB Image, 2×2 grid



800 × 600

Spectrogram image from GravitySpy training dataset v1.1.0*

- Timescale: 0.5, 1.0, 2.0, 4.0s
- Number of classes: 10 classes
- Number of used images: 921 test set data

*Data available on https://zenodo.org/records/1476551

(B) Prompt Text

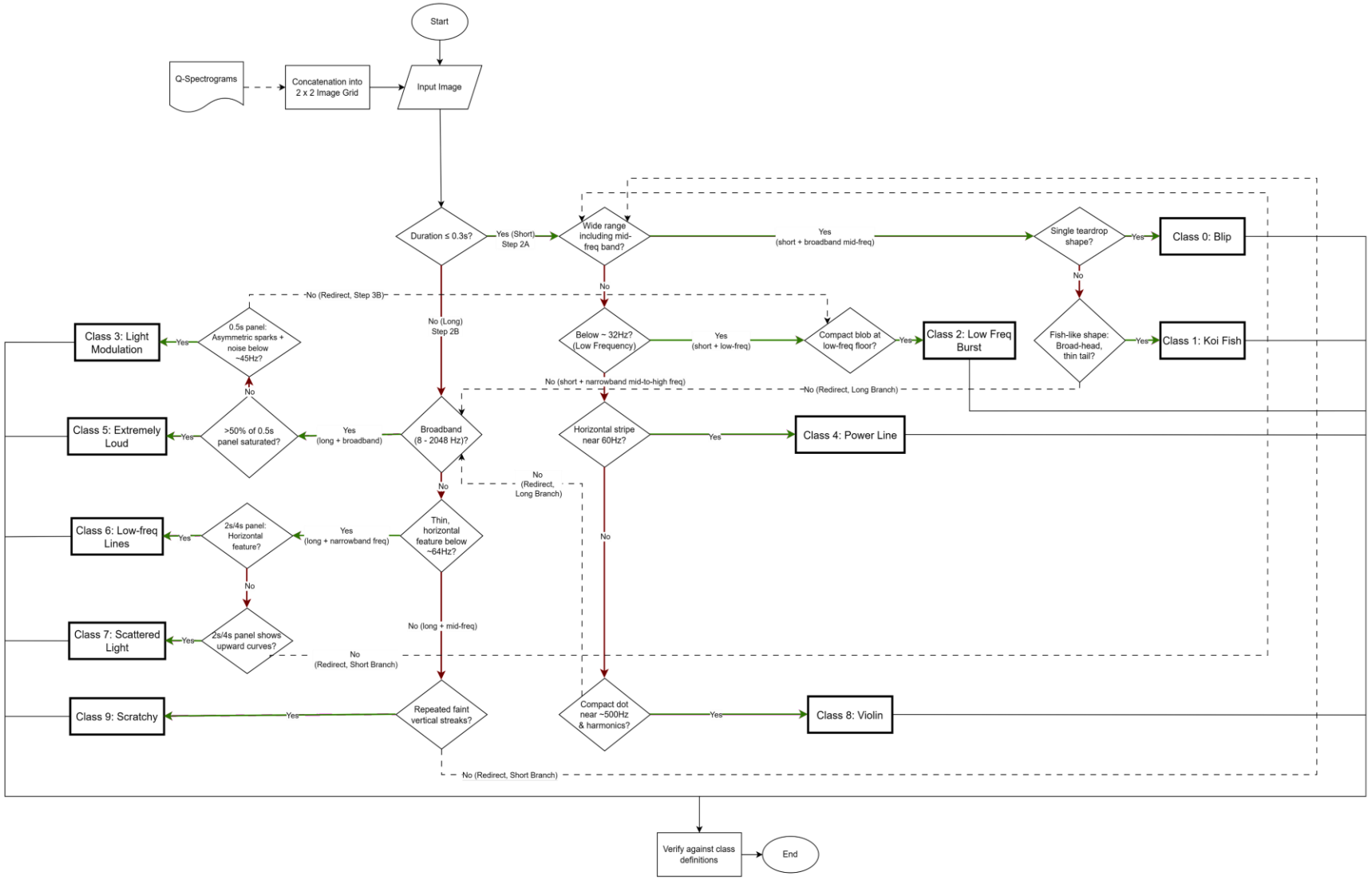
System Prompt: Defines how the model should behave, interpret visual features, and structure its outputs.

- Persona
- Input Image Format
- Spectrogram Axis Definitions
- Task Definitions
- Diagnostic Feature Guide

User Prompt: Provides task-specific content for each classification run

- Agent-specific Task Instruction
- Heuristic Guideline
- Output Format

Heuristic Guideline for Classifier Agent Prompt



Decision tree

Step1: Duration Gate
Step2: Frequency Gate
Step3: Shape Gate

Improved performance for single-agent setting

65.3%
(only with class definition)

⬇

71.6%
(adding heuristic guideline)

(C) Vision Language Model



OpenRouter

Selected Model: Gemma-4-31B-IT

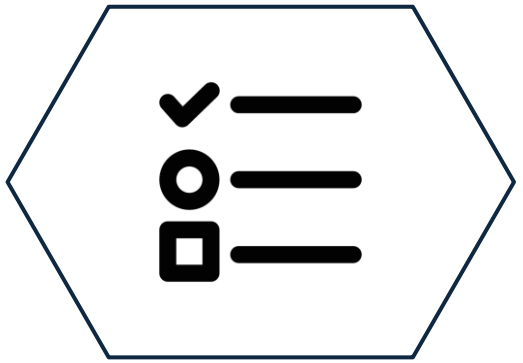
- Gemma is an open-weight multimodal model developed by Google DeepMind
- 31-billion-parameters with a 256K-token context window, handling text and image inputs
- Strong multimodal reasoning performance (vision-language benchmark MMMU Pro: 76.9%)

API Integration via OpenRouter

- Provides unified API access to multiple LLM/VLM providers via a single OpenAI-compatible endpoint
- Single API key and unified billing support reproducible and vendor-flexible experiment pipelines

Multi Agent Approach

Introducing Agents



Classifier Agent: Performs zero-shot classification

- Receives a merged spectrogram image and a heuristic classification prompt as input
- Outputs the predicted class as an integer label



Describer Agent: Generates a structured visual description of the spectrogram

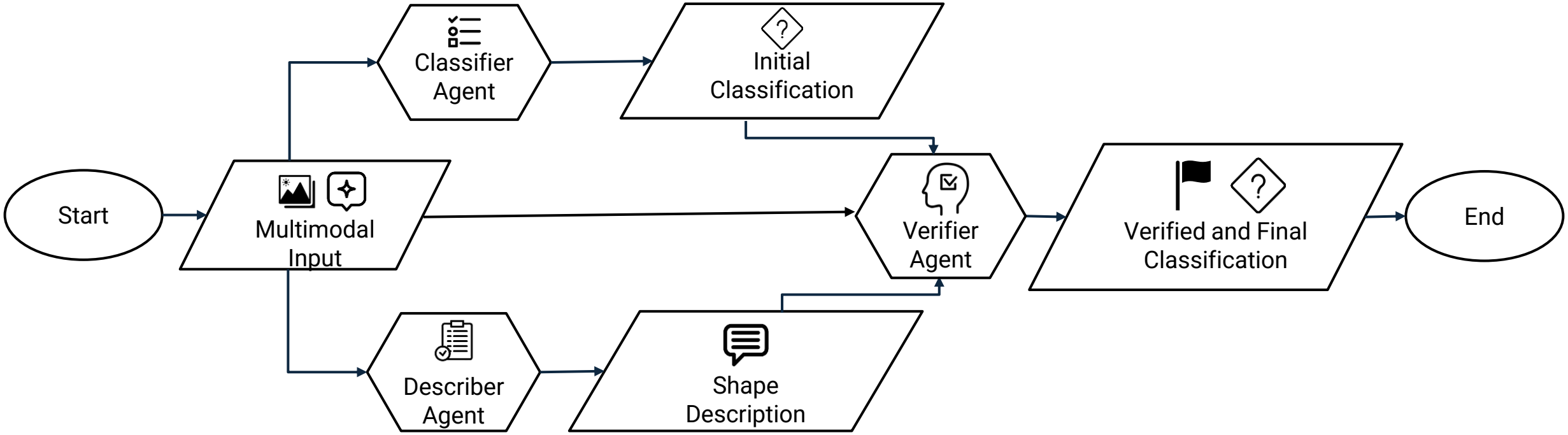
- Receives the same merged spectrogram image as the classifier agent, and runs in parallel
- Produces a text-based morphological summary that provides grounding for the classifier agent's decision



Verifier Agent: Reviews the consistency between the visual description and the classific ation output

- Jointly considers the input image, class definitions, and outputs from both preceding agents
- Identifies misalignments or hallucinated features between the describer's description and the classifier's label against the official class definitions
- Generates an agreement flag on the classifier's initial decision and proposes a revised class l abel

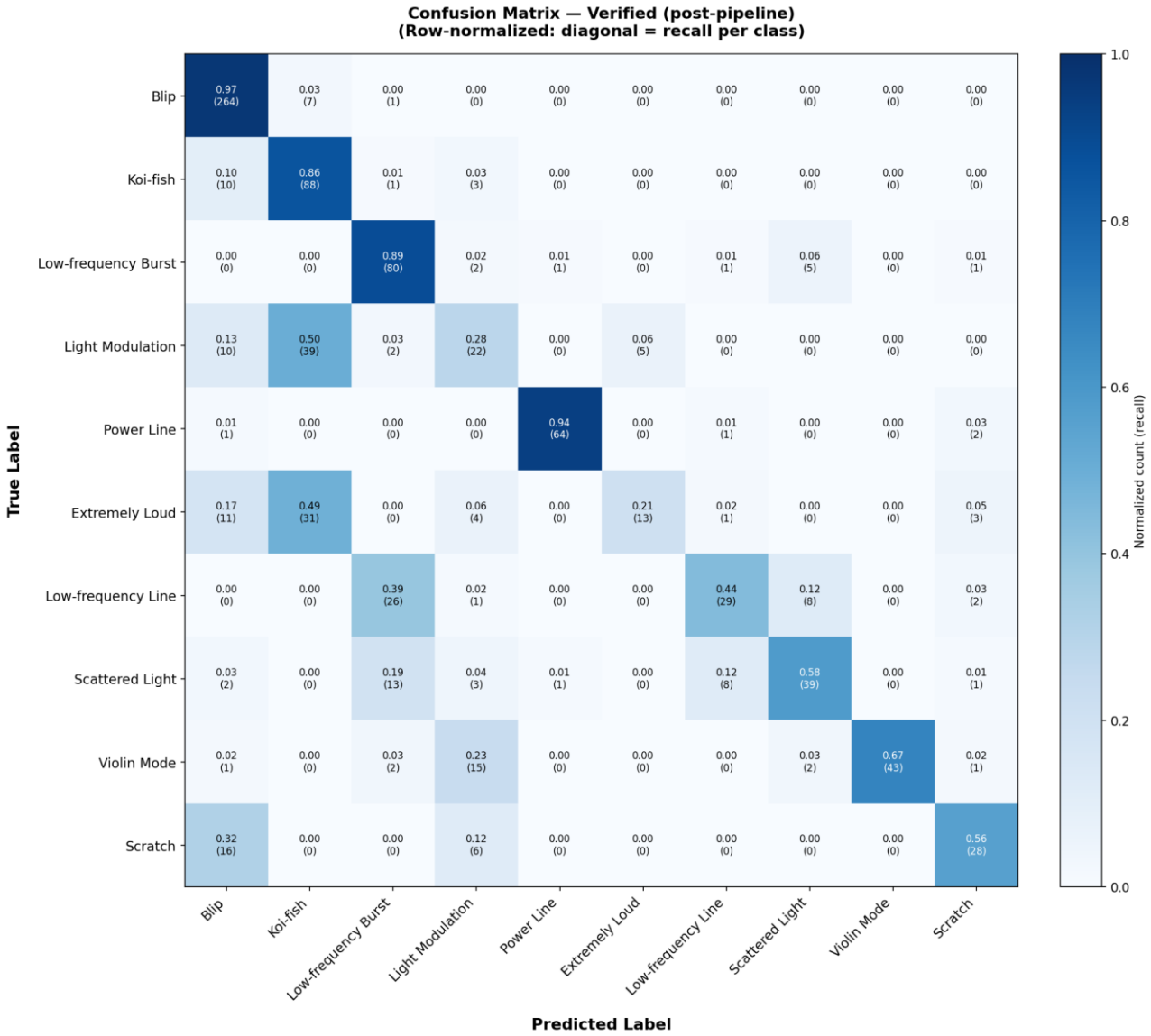
Multi Agent Pipeline



Agent	Input	Output
Classifier	Spectrogram input image + Prompt text (with heuristic guideline)	Integer label 0-9 (no desciption)
Describer	Spectrogram input image + Prompt text	Maximum 150 words free-text morphology description (no class name/ number)
Verifier	Classifier's integer label + Describer's text + Spectrogram input image + Prompt text (with class 0-9 definition)	Agree/Disagree flag + Final label 0-9

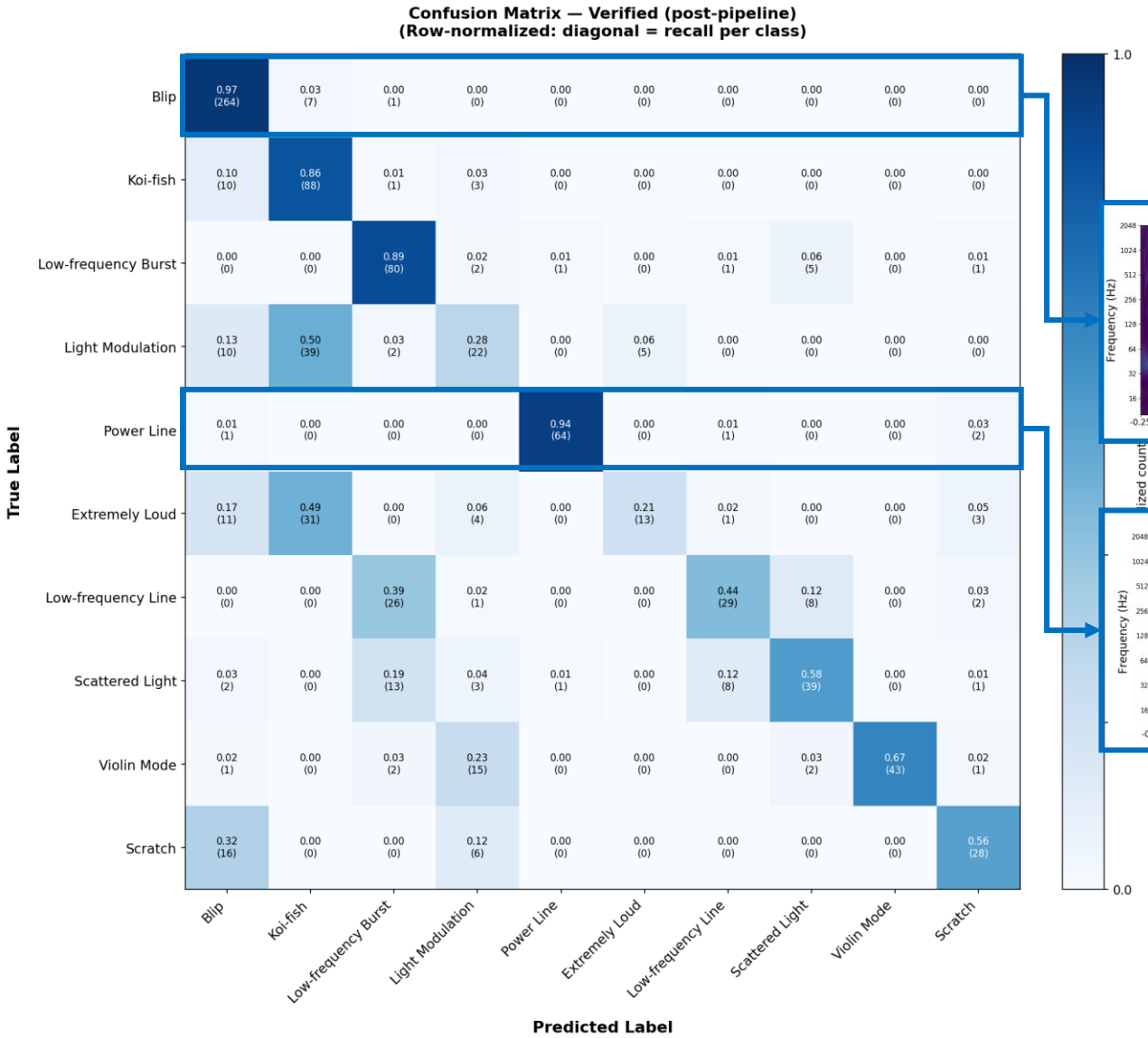
Classification Performance

Classification Performance per Class

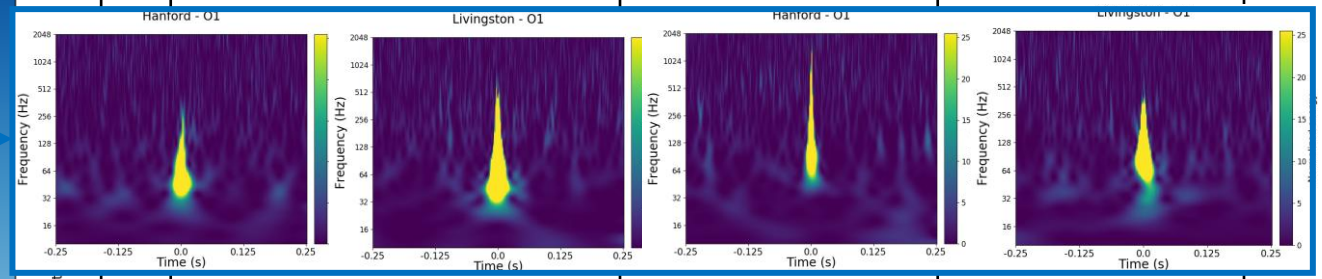


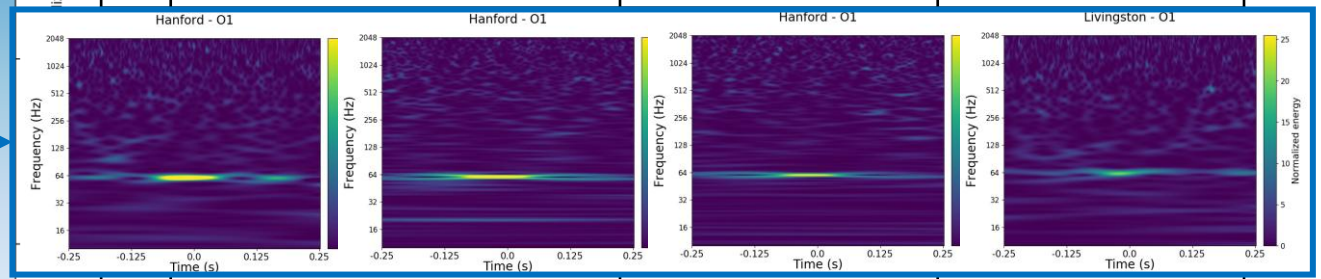
#	Class	Recall	Precision
0	Blip	0.971	0.838
1	Koi-fish	0.863	0.533
2	Low-frequency Burst	0.889	0.640
3	Light Modulation	0.282	0.393
4	Power Line	0.941	0.970
5	Extremely Loud	0.206	0.722
6	Low-frequency Line	0.439	0.725
7	Scattered Light	0.582	0.722
8	Violin Mode	0.672	1.000
9	Scratch	0.560	0.739

Classification Performance per Class

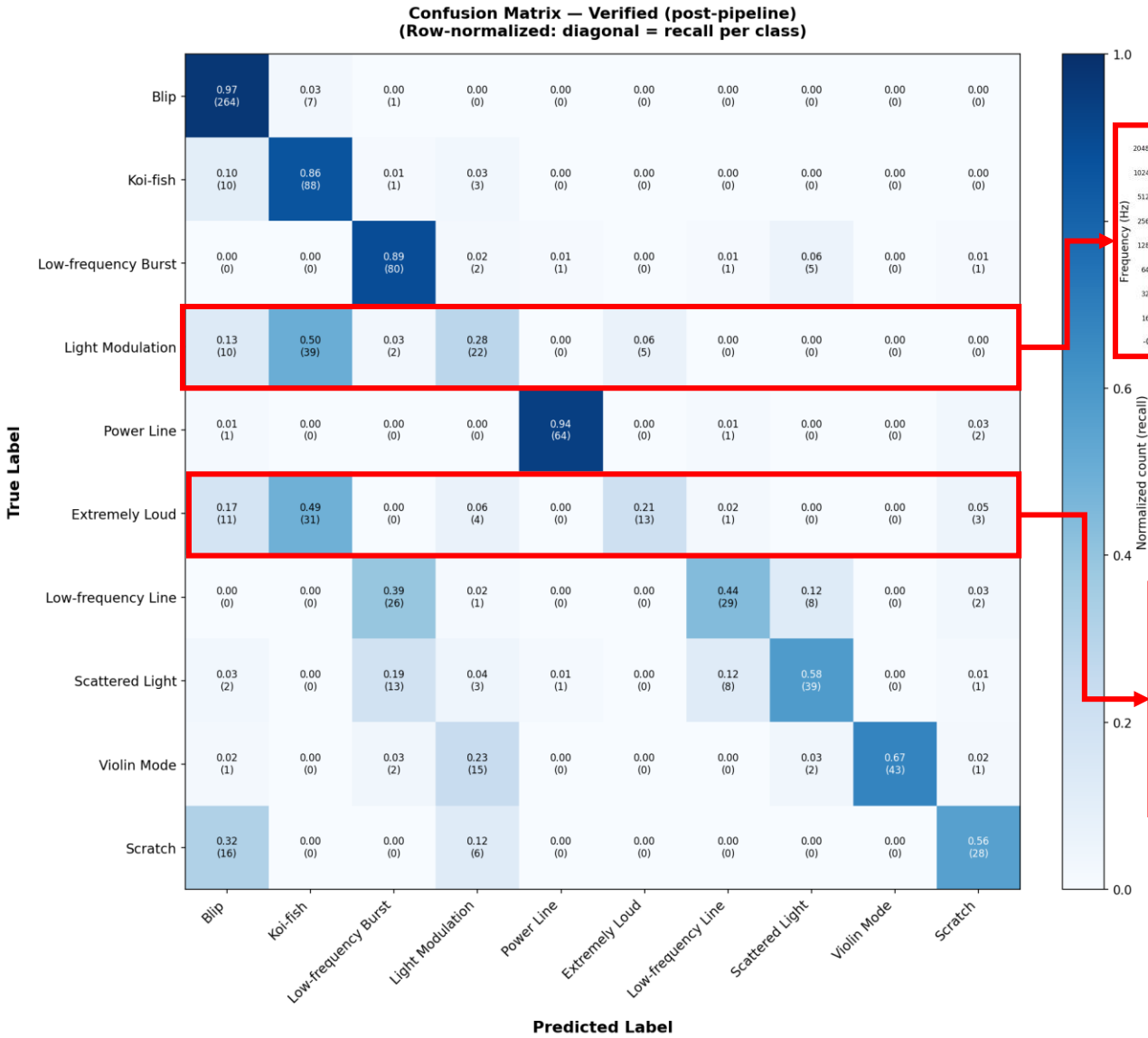


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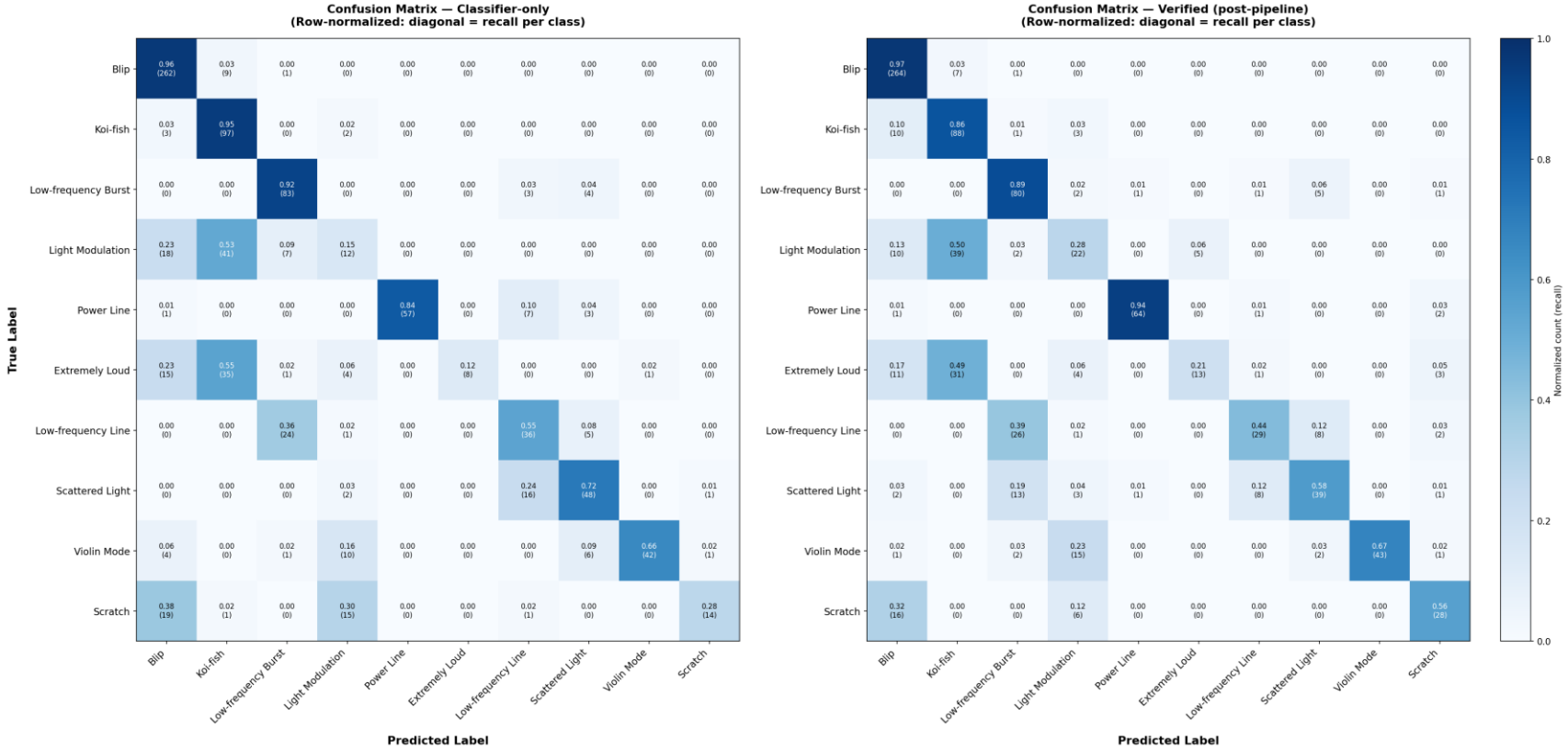


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#	Class	Recall	Precision
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How Verifier Improved Performance?

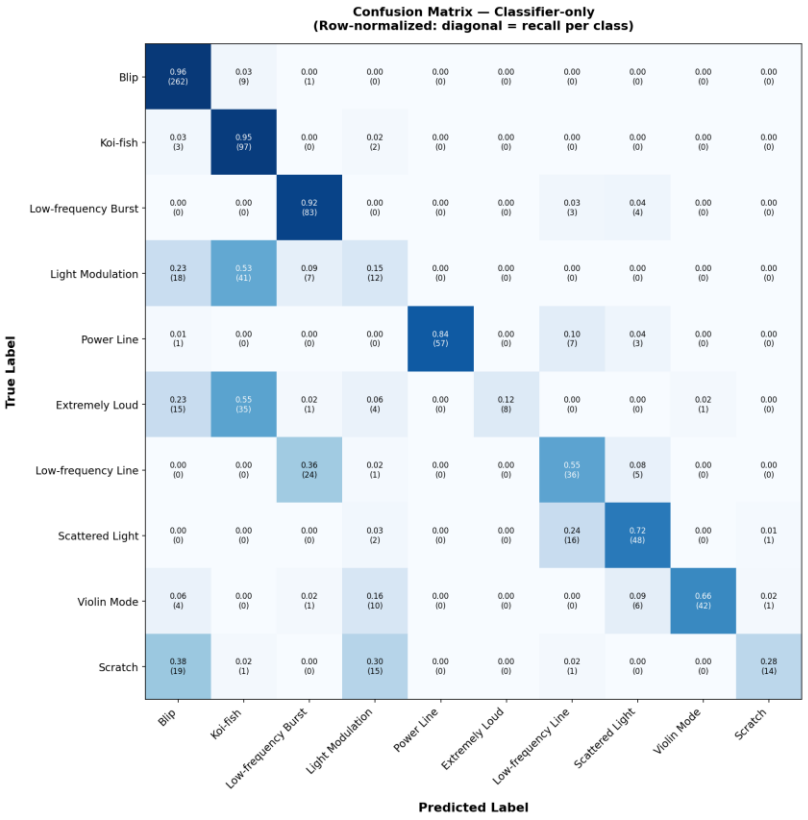


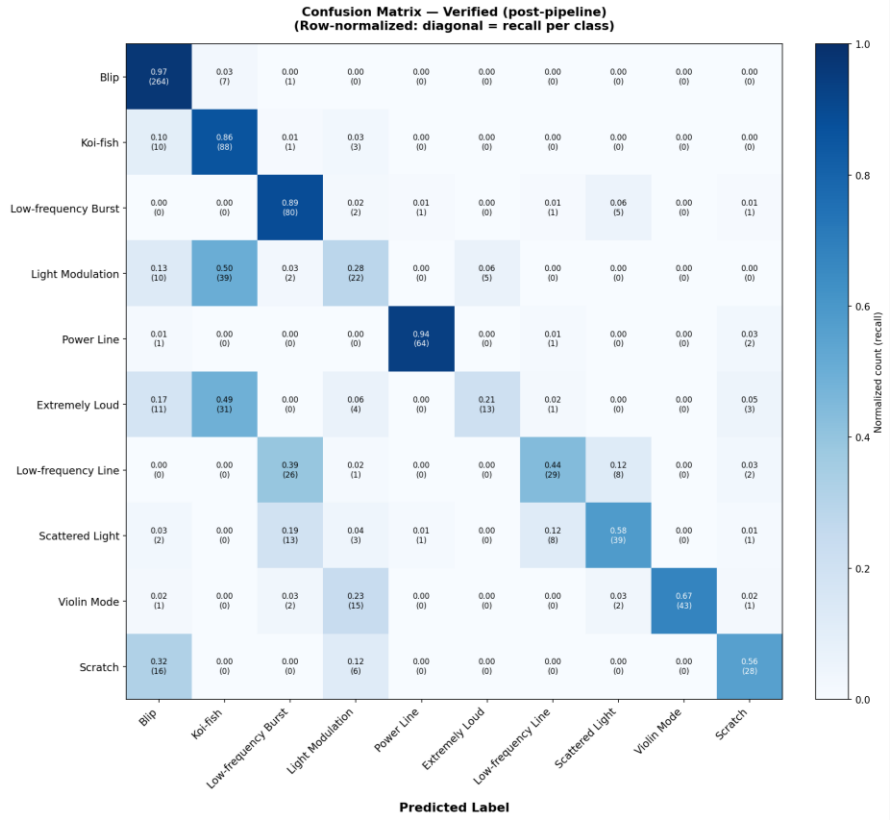
Agent	Accuracy	F1-Score
Classifier only	71.6 %	0.684
Multi Agents	72.8 %	0.709

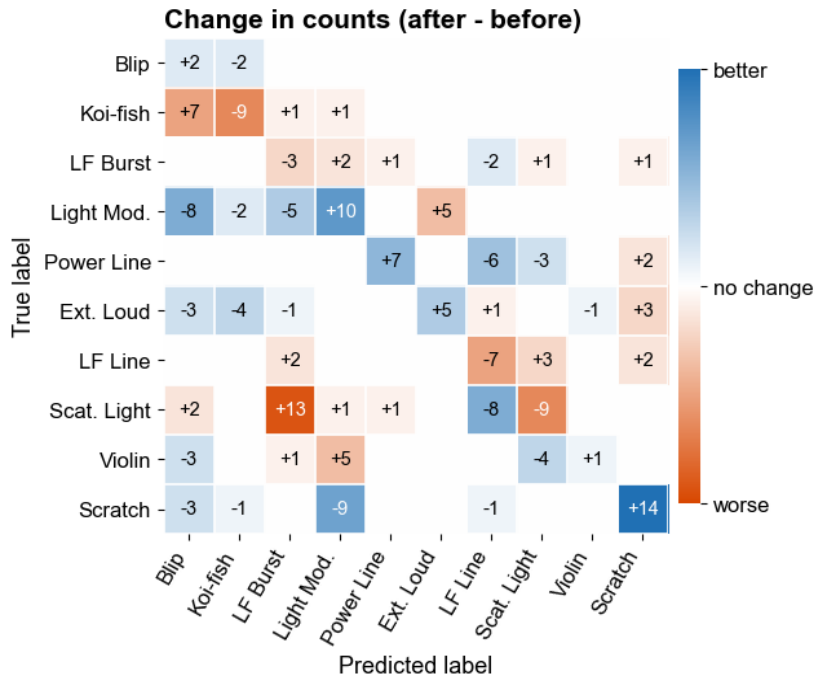
Verifier changed the label on 133 / 921 rows

- Corrected (wrong -> correct) 50
- Unsolved (still wrong) 44
- Regressed (correct -> wrong) 39

How Verifier Improved Performance?







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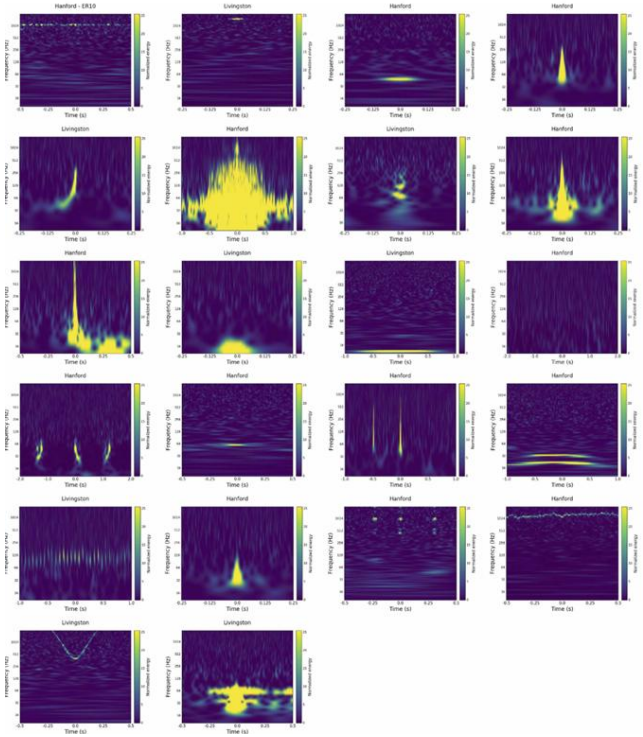
Net +11 correct

Limitation & Future Work

Limitations

- Evaluated on a single VLM (Gemma 4 31B)
- Heuristic prompt designed for 10 classes only
- Classes with high intra-class visual variability (e.g., Light Modulation, Extremely Loud) remain challenging regardless of classifying strategy

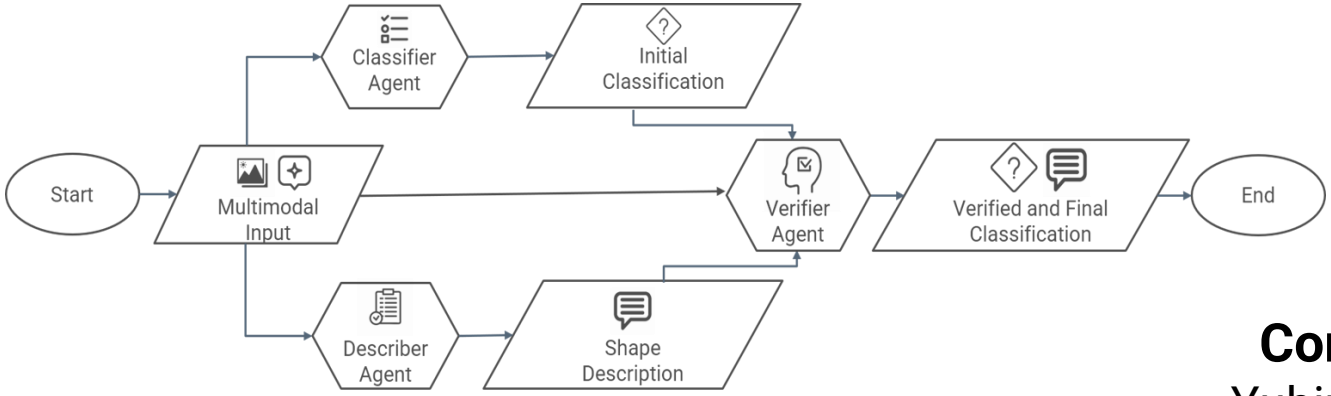
Future Work



- Extend multi-agent framework to the full 23-class Gravity Spy taxonomy
- Incorporate a None-of-the-Above (NoA) output option to flag candidate novel glitch morphologies
- Benchmark across multiple VLMs with evaluation metrics extended to inference latency and token usage

Summary

- Proposed a VLM-based GW glitch classification pipeline for zero-shot gravitational wave glitch classification, requiring no labeled training data or model fine-tuning.
- Introducing multi-agent pipeline improves overall accuracy from 71.6% (Classifier only) to 72.8% (Classifier + Describer + Verifier), and a multi-agent extension further enhances classification reliability and interpretability.
- Multi-agent VLM pipelines offer a scalable alternative to supervised machine-learning classifier for evolving detector environments.



Contact
Yubin Kim
National Institute for Mathematical Sciences
cosmos2718@ewha.ac.kr

Thank You for Listening

Contact
Yubin Kim
National Institute for Mathematical Sciences
cosmos2718@ewha.ac.kr

Backup Slides

10-Class Glitches Definition

- Machine learning for Gravity Spy: Glitch classification and dataset , S. Bahaadini et al. / Information Sciences 4 4 4 (2018) 172–186

ID	Class Name	Duration	Frequency	Description
0	**Blip**	Short	Mid	A single glitch usually have a duration of around 40 ms, frequencies between 30 and 500 Hz, and typically appear as a narrow, vertical and symmetric 'teardrop' shape in time-frequency domain.
1	**Koi fish**	Short	Mid, Low	These glitches are high-SNR broadband glitches. They typically occupy the frequency band ~20–1000Hz, and can be similar to Blip glitches, but they resemble a fish with the head at the low frequency end of the plot, pectoral fins around 30 Hz, and a thin tail around 500Hz.
2	**Low-frequency burst**	Short	Low	Loud, short-lived, low-frequency noise. Though multiple morphologies make up this category, they often resemble small humps with a nearly triangular shape growing from low frequency to a peak and then dying back down in a second. These are usually short-duration (~0.25s) transients between ~10–20Hz, bottom of the spectrogram.
3	**Light modulation**	Long	Mid, Low	The morphology of this glitch is not always the same, but it often looks like several bright spikes in close succession, often with low frequency noise at the same time. These transients are usually high SNR, with most of the noise content at 16–128Hz, but there may also be one or more broadband spikes.
4	**Power line**	Short	Low	These glitches usually look narrow in frequency, and last for about 0.2-0.5 seconds in time, centered around 60 Hz or one of its harmonics.
5	**Extremely Loud**	Long	High, Mid, Low	These broadband transients are characterised by very high SNR, often leading to the spectrograms appearing saturated. This is a catch-all category for glitches that result from a major disturbance in the detectors. They usually span much of the spectrogram and have extremely high energies.
6	**Low-frequency lines**	Long	Low	These appear mostly as flat lines, extending ~1.5–2s in time and usually below ~20Hz.
7	**Scattered Light**	Long	Low	They are often low-frequency, long duration, humpy glitches that look like one or several curved lines stacked on top of each other. They appears as long-duration (~2.0–2.5s) arches in the spectrograms.
8	**Violin Mode**	Short	High	This category appears short and dot-like, and occurs at 500 Hz and multiples (harmonics) of this frequency.
9	**Scratchy**	Long	High, Mid	A series of short-duration repeating glitches (often with 10-30 glitches per second) often lasting several seconds. These appear as a series of sharp peaks at intermediate frequencies ~60–250Hz. There may be ~10–30 peaks per second.

Performance Evaluation Parameter

Precision

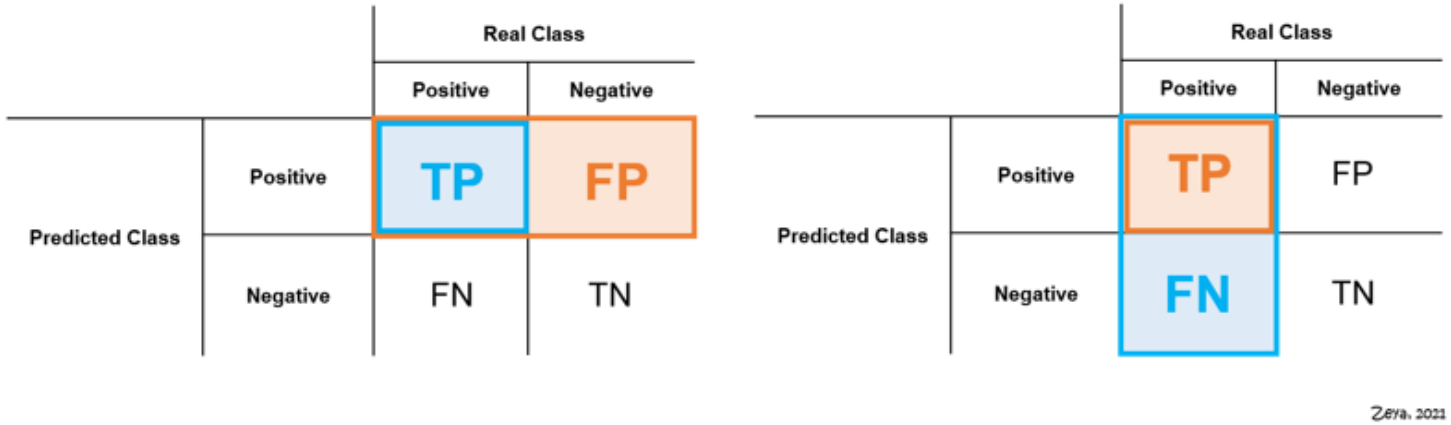
Of all positive predictions, how many are really positive?

$$\frac{TP}{TP + FP}$$

Recall

Of all real positive cases, how many are predicted positive?

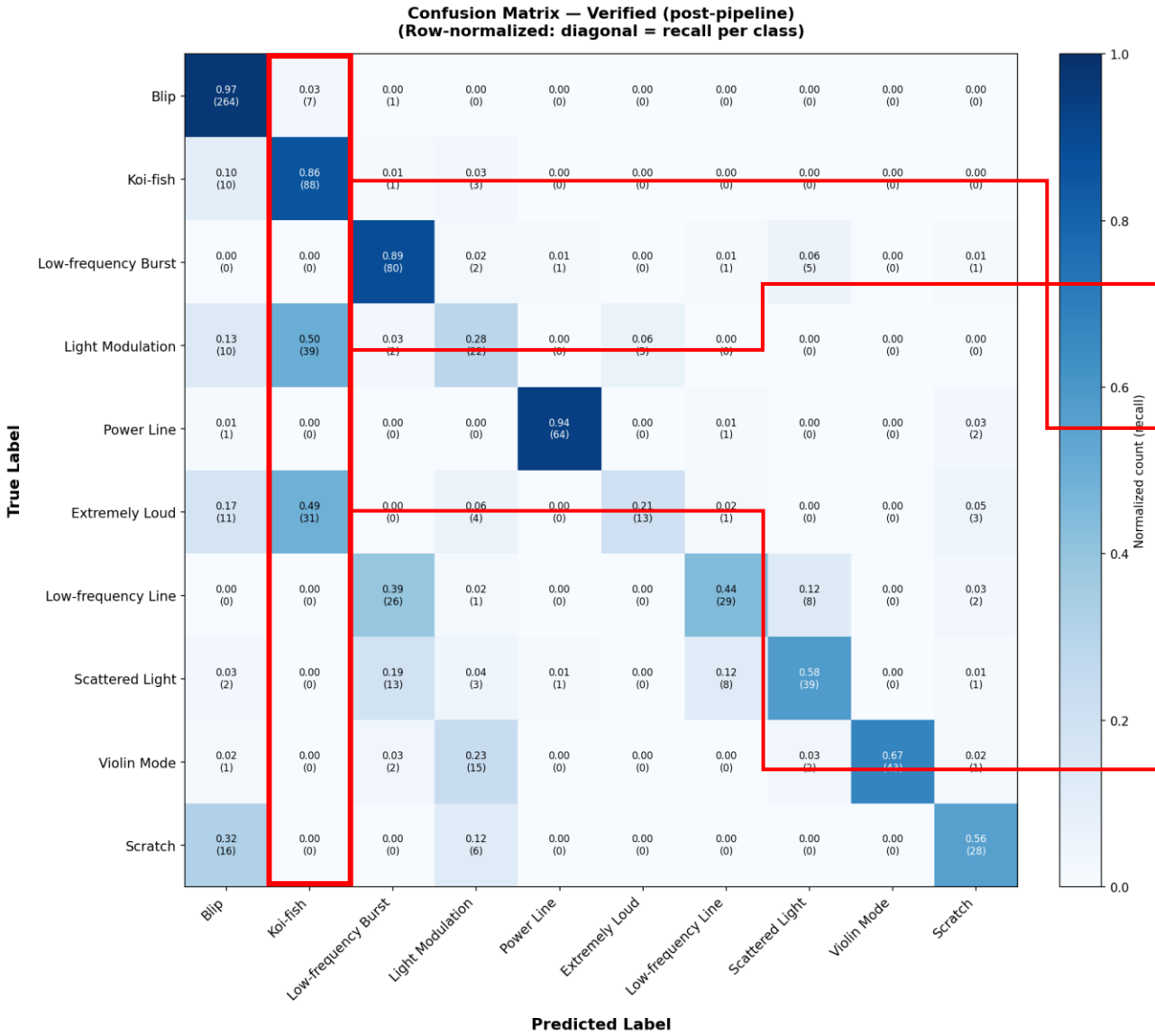
$$\frac{TP}{TP + FN}$$



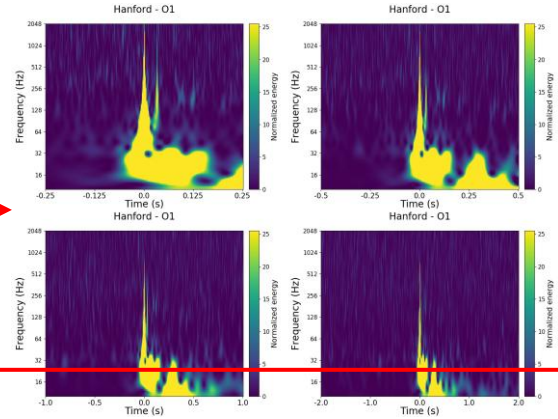
$$\text{F1 Score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Accuracy:** Proportion of all samples correctly classified across all classes.
- **Precision:** Of all samples predicted as a given class, the fraction that truly belong to it.
- **Recall:** Of all true samples in a given class, the fraction the model correctly identified.
- **Weighted F1-Score:** Harmonic mean of precision and recall, averaged across classes weighted by their sample count — accounts for class imbalance.

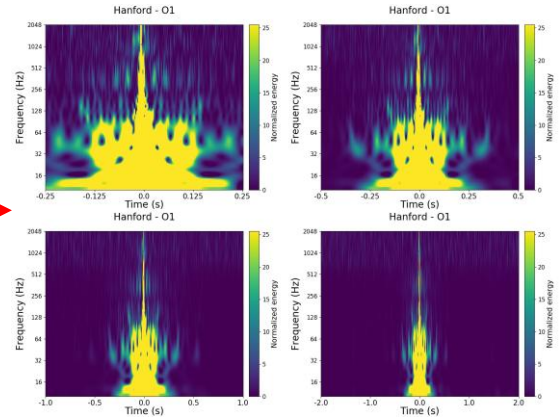
Misclassification Analysis



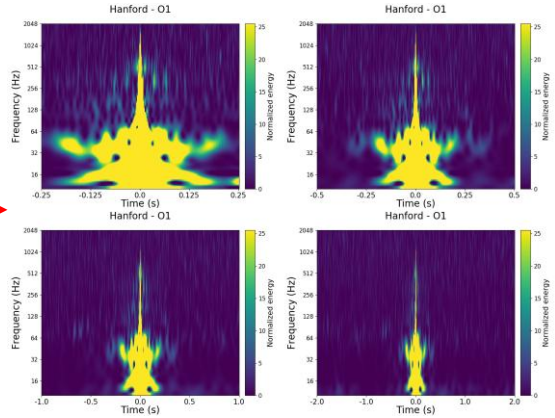
True Label: 3 (Light Modulation)
Predicted Label: 1 (Koi Fish)



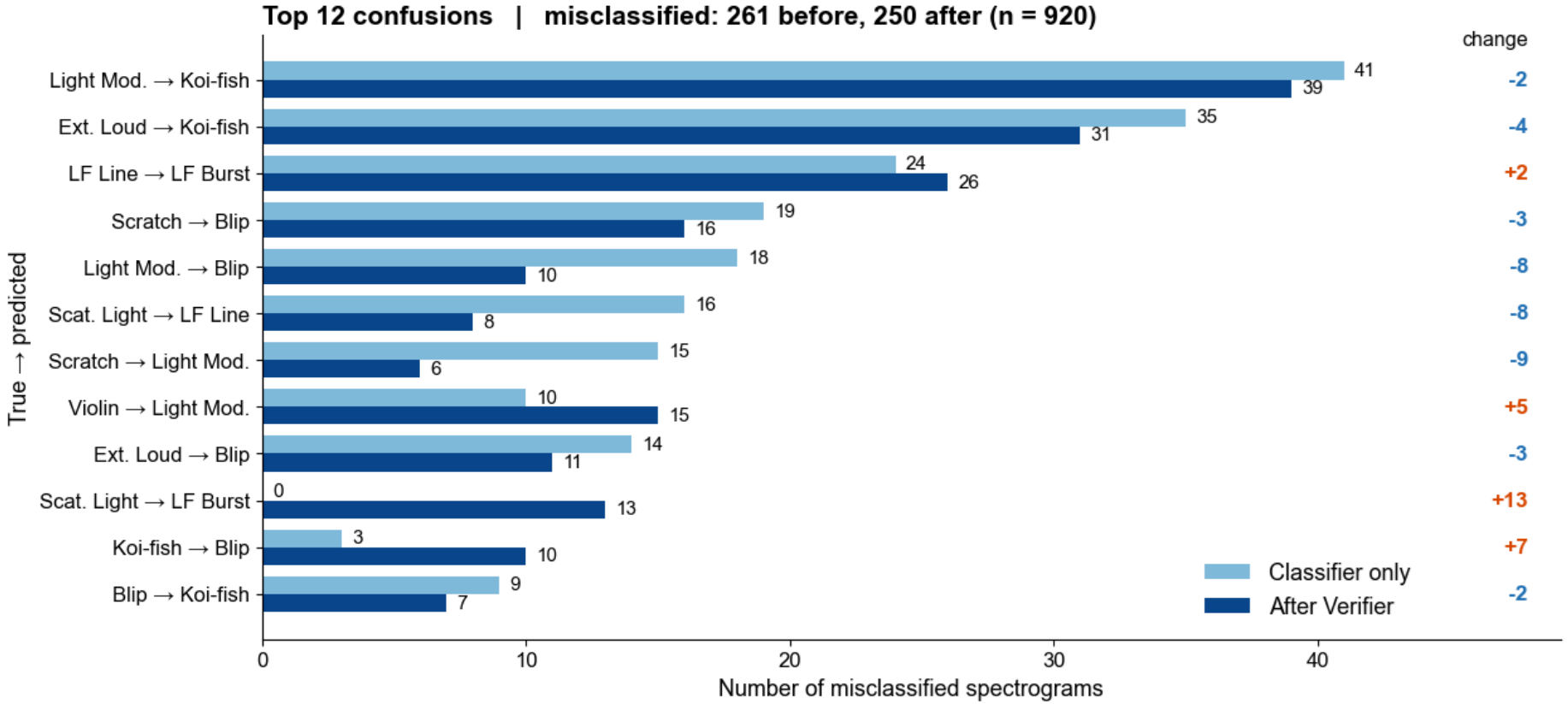
True Label: 5 (Extremely Loud)
Predicted Label: 1 (Koi Fish)



True Label: 1 (Koi Fish)
Predicted Label: 1 (Koi Fish)



Misclassification Analysis: Top Confusion Pairs



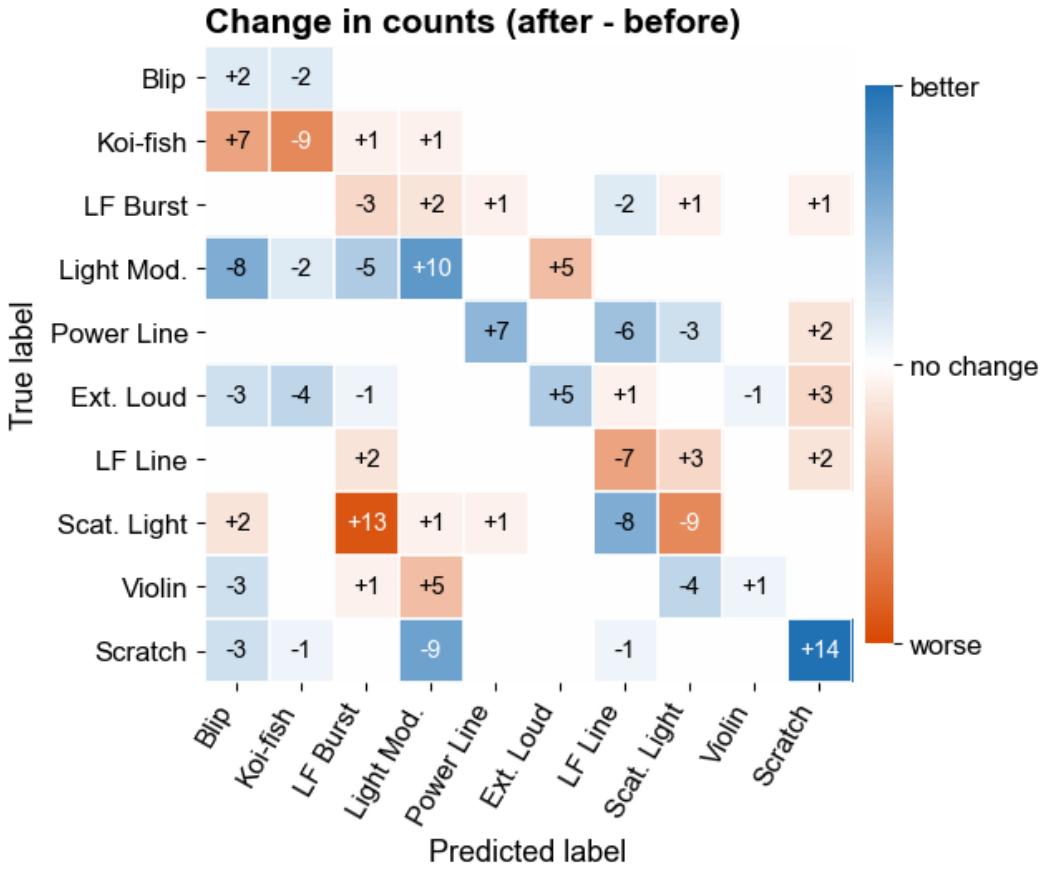
2 pairs, 28% of errors
Light Mod. and Ext. Loud → Koi-fish

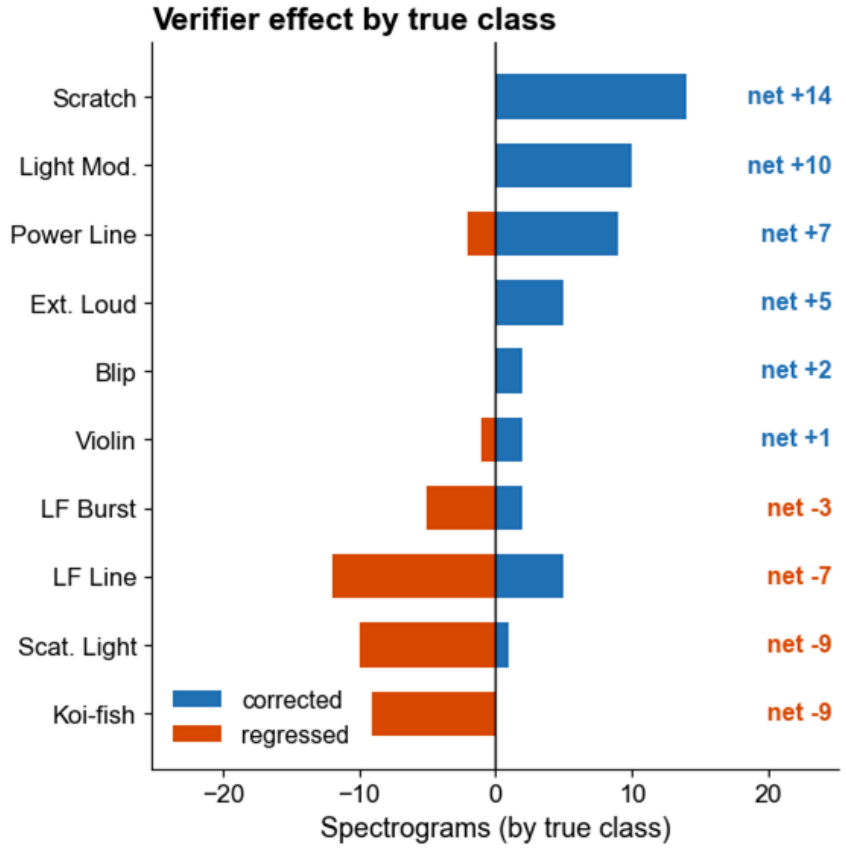
Short look-alikes collapse[1]
into Koi-fish and Blip

Top 2 barely moved
Verifier effect: -2 and -4 spectrograms

[1] Blips are hard to separate from other short-impulse transients: Cabero et al., Class. Quantum Grav. 36, 155010 (2019)

Misclassification Analysis: Verifier Agent's Performance



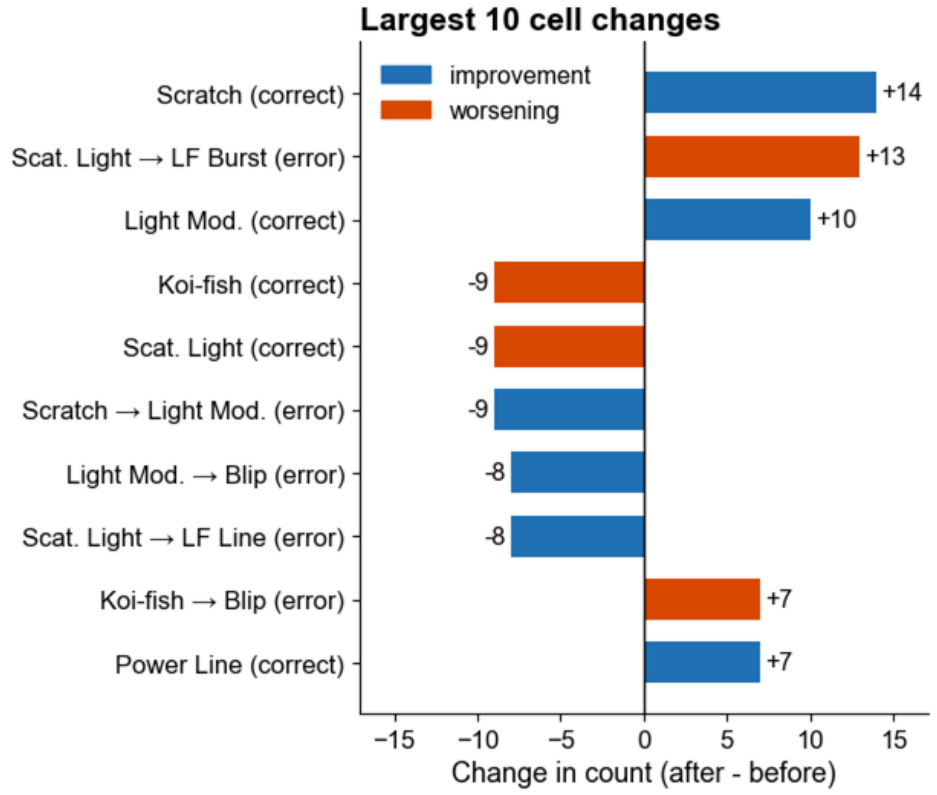


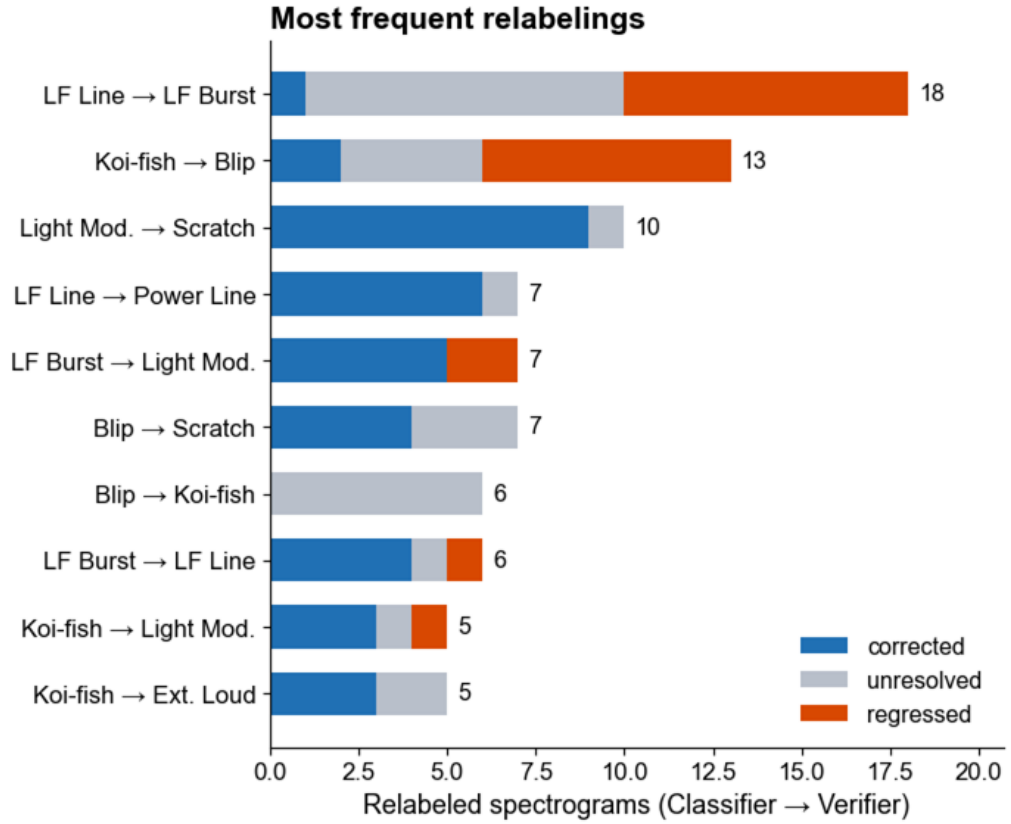
Net +11 correct
50 corrected, 44 unresolved, 39 regressed

Best: Light Mod. → Scratch
9 of 10 relabelings corrected

Worst: LF Line → LF Burst
1 of 18 corrected, 8 regressed

Misclassification Analysis: Verifier Agent's Performance





Scratch +14, Light Mod. +10
correct predictions gained

Koi-fish -9, Scat. Light -9
correct predictions lost

Some errors move sideways
Scat. Light errors: LF Line -8, LF Burst +13

Token Usage

Agent	N calls	Input tokens / call	Output tokens / call	Total tokens / call	Latency (s), mean ± SD	Total tokens (sum)	Est. cost (USD)
Classifier	921	2,366.0 ± 0.0	2.0 ± 0.0	2,368.0 ± 0.0	2.63 ± 1.31	2,180,927	0.284
Describer	921	771.9 ± 0.3	169.9 ± 12.4	941.9 ± 12.4	9.87 ± 7.27	867,449	0.152
Verifier	921	1,566.6 ± 12.5	4.0 ± 0.1	1,570.6 ± 12.5	1.87 ± 1.66	1,446,504	0.189
Pipeline (per image)	2763	—	—	4,880.4 ± 24.9	14.37 ± 7.66	4,494,880	0.625

Model : google/gemma-4-31b-it Image detail : high Rows (images) : 921 Agents in chain : 1